

Multiple Choice

1. **Answer:** D

Options: A) the softer rocky material of the mantle.; B) the lava that comes out of volcanoes.; C) material between the crust and the mantle.; D) the rigid rocky material of the crust and uppermost portion of the mantle.

Note: Correct option: D - the rigid rocky material of the crust and uppermost portion of the mantle.

2. **Answer:** D

Options: A) From the layering of materials within the Earth.; B) From fossils of ancient life.; C) From the cratering history of Earth's surface.; D) From radioactive dating of rocks and meteorites.

Note: Correct option: D - From radioactive dating of rocks and meteorites.

3. **Answer:** A

Options: A) third quarter.; B) waning crescent.; C) waxing crescent.; D) full.

Note: Correct option: A - third quarter.

4. **Answer:** D

Options: A) It would be harder since the truck is heavier on Mars.; B) It would be easier since the truck is lighter on Mars.; C) It would be harder since the truck is lighter on Mars.; D) It would be the same no matter where you are.

Note: Correct option: D - It would be the same no matter where you are.

5. **Answer:** D

Options: A) hydrogen compounds (H₂O CH₄ NH₃) light gases (H He) metals rocks; B) hydrogen compounds (H₂O CH₄ NH₃) light gases (H He) rocks metals; C) light gases (H He) hydrogen compounds (H₂O CH₄ NH₃) metals rocks; D) light gases (H He) hydrogen compounds (H₂O CH₄ NH₃) rocks metals

Note: Correct option: D - light gases (H He) hydrogen compounds (H₂O CH₄ NH₃) rocks metals

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'From the dismantling of small moons by impacts'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'Barnard's Star'.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Orion. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** The force of attraction between any two objects decreases with the square of the distance between their centers.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.